MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) Regular EXAMINATION October 2025
Third SEMESTER
Minor Course Aquaculture- MG3DSCAQC202
ANATOMY AND PHYSIOLOGY OF FIN FISHES AND SHELL FISHES
Question paper code :

Part A

Very Short Answer Questions
Answer all questions
Each question carries 1 marks

1. Give an example of skin of fishes performing respiratory function
   Eel
2. How do mature ovaries appear in breeding season?

The ovaries become large, swollen, and occupy most of the body cavity, vascular, and highly turgid due to the presence of fully developed ripe ova (eggs). Color commonly yellowish, orange, or reddish. The eggs are often visible through the thin ovarian wall.

3. What is Mermaid's purse of elasmobranch?

The Mermaid's purse in elasmobranchs refers to the tough, leathery egg case that encloses and protects the developing embryo in oviparous sharks and skates.

4. Name the two portal systems in fish?

1. Hepatic portal system 2. Renal portal system

5. Where are the testes located in shrimp?

In shrimp, the testes are located in the dorsal region of the cephalothorax, just beneath the carapace and above the hepatopancreas and stomach.

6. What is the primary function of the esophagus in finfishes?

The primary function of the esophagus in finfishes is to transport food from the buccal cavity to the stomach.

7. Which morphometric paramater in fishes is most commonly used for accurate growth analysis?

The total length (TL) is the most commonly used because it is easy to record accurately and correlates well with body weight.

8. Name one method to study spawning season.

Gonado somatic index

9. What are blood sinuses in the bivalve circulatory system?

In bivalves, blood sinuses are irregular, open spaces or cavities through which blood flows freely instead of being confined to closed vessels.

10.In crabs at which maturity stage is the vas deferens occupy the entire body cavity?

Fully mature (ripe) stage of gonadal development.

Part B

Short Answer Questions
Answer any 5 questions
Each question carries 6 marks

11. Distinguish between Potamodromous and Oceanodromous migration of fishes?

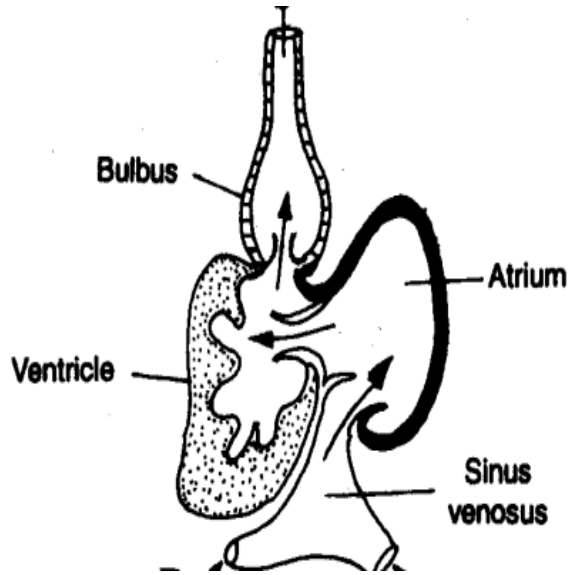
Potamodromous migration: Migration that occurs entirely within fresh water

Oceanodromous migration: Migration that occurs entirely within marine waters.
Purpose of Migration : Both types involve movement for spawning, feeding, or overwintering.
Examples : Potamodromous: *Labeo rohita, Catla catla,* Oceanodromous: *Tuna, Mackerel*.
12. Discuss the excretion of nitrogenous waste in finfishes.

Major waste: Ammonia, Mode: Ammonotelic, Organs: Gills and kidneys, Explain Mechanism: Diffusion and active transport Urea, uric acid in some species, Significance: Adaptation to aquatic life and salinity regulation

13. Explain the different types of social behavior in fishes with suitable examples.
Shoal: A shoal is a loose group of fishes that stay together mainly for social reasons such as protection, feeding, or mating, but without synchronized movement or uniform direction.
School: A school is a highly organized and synchronized group of fishes that swim together in the same direction, maintaining uniform spacing and movement.
Colony : A colony refers to a group of fishes that live together in a defined area for nesting, breeding, or shelter but are not necessarily moving together.
Cluster : A cluster is a compact aggregation of fishes formed temporarily, usually as a defensive or resting behavior.
Briefly mention the properties and functions
14. Describe the structure of the heart of teleosts, with a suitable diagram?

Two chambered, venous heart, Explain sinus venosus, atrium (Auricle), ventricle, bulbus arteriosus.

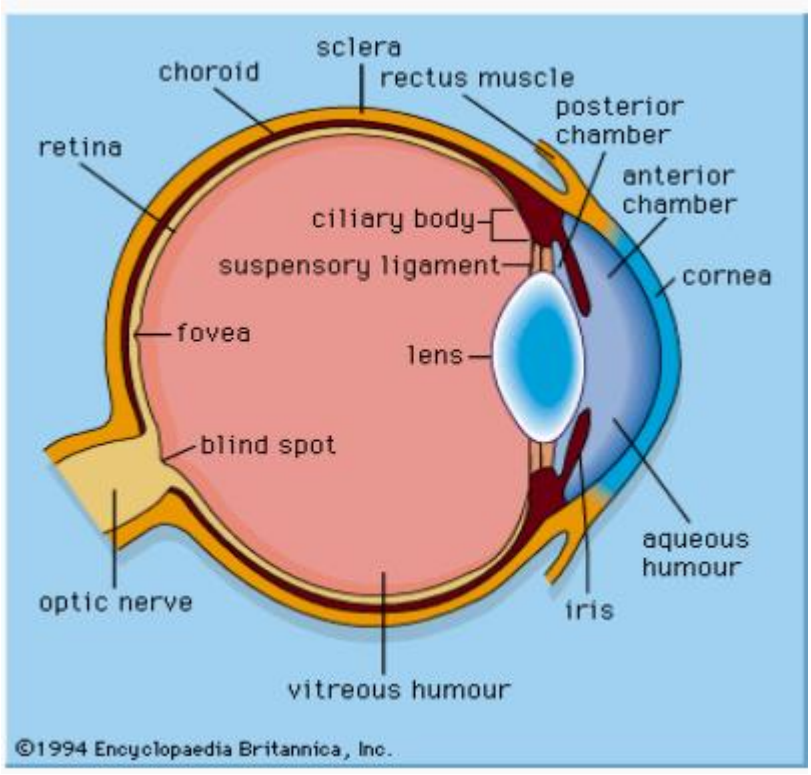


15.Write a note on the natural food of bivalves. Describe the feeding mechanism in bivalves

Natural food: Phytoplankton (diatoms, dinoflagellates, green algae), Zooplankton (protozoans, small larvae), Organic particles and bacteria. Bivalves are ciliary-mucus feeders using gills (ctenidia) and labial palps. They filter water to trap microscopic food particles. Explain process of feeding : water current, filtration, inhalent and exhalent siphons

16.Write a note on swim bladder in teleost fishes
Swim bladder or air bladder or gas bladder, position, physostomous, physoclistous, retia mirabilia, red bodies, anterior gas secreting chamber and a posterior gas absorbing chamber, ovales, diaphragm, red glands, pneumatic duct, Functions of swim bladder, respiratory, sound production, auditory, sensory, hydrostatic organ, explain briefly
17.Draw the structure of fish eye and label any 4 parts

Draw the Diagram. Label any 4 of these parts: Retina, choroid, sclera, optic nerve, aqueous humour, vitreous humour, iris, lens, cornea, blind spot, rectus muscle, ciliary body, suspensory ligament



18. Describe the role of excretion in maintaining ionic balance in both shrimps and bivalves.

Excretory organs: antennal (green) glands located at the base of the antennae.

These glands help remove nitrogenous waste (mainly ammonia) and regulate ions like Na^+, K^+, and Cl^-.

In marine shrimps, the glands help eliminate excess salts absorbed from seawater.

In freshwater shrimps, they help conserve salts and expel excess water by producing dilute urine.
Excretion and Ionic Regulation in Bivalves

Bivalves : excretory organs are paired kidneys (nephridia) located near the pericardium.

They excrete nitrogenous waste mainly as ammonia.

Kidneys also help regulate ionic concentration in the haemolymph.

Marine bivalves maintain osmotic balance by adjusting ionic exchange across gills and nephridia.

Freshwater bivalves produce dilute urine to expel excess water while retaining salts.
19. Explain the mechanism of respiration in Teleosts?

Gills are located in the gill chambers on each side of the pharynx, protected by operculum. Each gill consists of: Gill arches, Gill filaments, Secondary lamellae (site of gas exchange)Rich blood supply and thin epithelium allow efficient diffusion of gases. The mouth opens and the buccal cavity floor lowers, increasing its volume. Water enters the buccal cavity. The opercular cavity remains closed during this phase. The mouth closes, and the buccal floor rises, decreasing cavity volume. Opercular valves open, allowing water to flow over the gills and out through the opercular openings. Gas exchange occurs as: O_2 diffuses from water into blood, CO_2 diffuses from blood into water. Blood in the secondary lamellae flows in the opposite direction to the water current. This counter-current mechanism maintains a concentration gradient, maximizing oxygen absorption efficiency.

Part C

Essay Questions

Answer any 1 questions

Each question carries 10 marks

20.. Describe the role of the stomach, gastric mill, hepatopancreas, midgut, and hindgut in the digestion and absorption of food in shrimp.

Stomach: Divided into: Cardiac and pyloric chambers

Function: Mechanical digestion and filtration, Cardiac stomach: Receives food, performs grinding, Pyloric stomach: Filters fine particles to midgut

Gastric Mill: Location: Inside cardiac stomach, Structure: Three chitinous ossicles (tooth-like plates), Function: Mechanical trituration of food, Ensures only small particles pass to pyloric region.

Hepatopancreas: Combined liver + pancreas, Function: Secretes digestive enzymes – protease, amylase, lipase, Also involved in Absorption, storage, and metabolism

Midgut: Receives: Enzymes from hepatopancreas, Function: Chemical digestion and absorption of nutrients, Absorbs Amino acids, sugars, fatty acids, glycerol, Adaptation: Presence of microvilli for increased surface area

Hindgut: Function: Water and ion reabsorption, Forms: Fecal pellets for elimination, Lined with: Chitin for protection

21. Describe the structure and function of major sense organs (eye, ear, lateral line and olfactory organ) in fishes .

Eye (Organ of Vision)

Structure: Spherical lens – almost perfectly round for underwater vision, Cornea – thin and flat; minor role in refraction, Iris – regulates light entry., Retina – contains rods (dim light) and cones (bright light, color), Choroid gland – rich in blood vessels; supplies oxygen to retina. No eyelids; eyes adapted for aquatic medium.

Function: Provides monocular and binocular vision, Rods – aid in night vision; cones – detect colors, Helps in locating prey, avoiding predators, and navigation.

2. Ear (Organ of Hearing and Equilibrium)

Structure:Internal ear only; no external or middle ear, Located in otic capsule of skull, Contains three semicircular canals (balance) and otolith organs (hearing)., Otoliths – calcium carbonate structures that detect vibrations. Connected to swim bladder by Weberian ossicles in some fishes (e.g., carps).

Function: Hearing – detects sound vibrations from water. Equilibrium – maintains body position and orientation.

3. Lateral Line

Structure: A line of neuromasts (sensory cells) arranged in canals on head and body sides.Neuromasts contain hair cells enclosed in a gelatinous cupula. Open to the exterior through pores.

Function: Detects vibrations, water currents, and movement around the body.

Helps in schooling, prey detection, and obstacle avoidance.

Functions effectively even in dark or turbid water.

4. Olfactory Organ (Organ of Smell)

Structure: Pair of olfactory sacs or nares located on snout. Each sac lined with olfactory epithelium bearing sensory and ciliated cells. Water enters through anterior nostril and exits through posterior nostril.

Function: Detects chemical stimuli (smell and taste) from water. Important for food detection, predator avoidance, and spawning migration. Highly developed in migratory fishes like salmon and eel

22. Describe various types of growth in finfishes and explain their importance in fisheries biology

Growth – permanent increase in size, weight, or volume of a fish over time.

Caused by cell division, enlargement, and tissue accumulation.

Types of Growth in Finfishes

A. Absolute Growth

Measures total increase in length or weight over time.

Formula: $\frac{W2-W1}{t2-t1}$.

Indicates overall change in body size.

B. Relative Growth

Describes growth of one body part relative to another or to body size.

relative growth = $\frac{W2-W1}{W1}$

Relative Growth rate is expressed as $\frac{W2-W1}{W1(t2-t1)}$

C. Isometric Growth

All parts grow at the same rate, maintaining same body proportions.

Cube Law

In length–weight relationship, b = 3.

Indicates balanced and uniform growth.

D. Allometric Growth

Unequal growth of body parts.

Positive allometry: one part grows faster ($b > 3$).

Negative allometry: one part grows slower ($b < 3$).

Importance of Growth Studies in Fisheries Biology

Age and Growth Estimation – determines fish age using scales, otoliths, etc.

Stock Assessment – helps estimate population and maximum sustainable yield (MSY).

Fishery Management – identifies optimum size for capture and recruitment.

Aquaculture Practices – selection of fast-growing species, optimizing feed and culture conditions.

Environmental Indicator – growth rate reflects water quality and habitat conditions.

****